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Professional Preparation

Universidad Nacional Autónoma de México	Physics	B.Sc., 1981
Cambridge University	Biophysics	Ph.D., 1985
Harvard University	Biophysics	1985-1989

Professional Experience

Undergraduate research with Dr. I. Ortega-Blake, Universidad Nacional Autónoma de México, 1980-1981
Graduate student with Drs. A.C. Bloomer and A. Klug. MRC Laboratory of Molecular Biology, University of Cambridge. 1981 - 1984
Post-doctoral Fellow with Prof. S.C. Harrison. Department of Biochemistry and Molecular Biology, Harvard University. 1985 - 1989
Assistant Professor. Department of Biochemistry, Molecular Biology and Cell Biology, Northwestern University, Evanston, IL. 1989 - 1994
Associate Professor. Department of Biochemistry, Molecular Biology and Cell Biology, Northwestern University, Evanston, IL. 1994 – 2000
Professor. Department of Biochemistry, Molecular Biology and Cell Biology, Northwestern University, Evanston, IL. 2000 -

Awards

University Achievement Award, Universidad Nacional Autónoma de México, 1982.
Overseas Research Scholarship (U.K.) 1981 - 1984.
Scholarship from Universidad Nacional Autónoma de México, 1981 - 1984.
Damon Runyon-Walter Winchell Cancer Fund Research Fellowship, June 1985 - June 1988
Owen L. Coon Chair in Molecular Biology, Northwestern University, 2002 - 2004

Service

Member. BBCA Study Section, NIH-NIGMS, 1996 – 2000
Director. Center for Structural Biology, Northwestern University
Director. Robert H. Lurie Comprehensive Cancer Center Structural Biology Facility, Feinberg School of Medicine, Northwestern University
Member, Scientific Advisory Committee, Stanford Synchrotron Radiation Laboratory, 2007-2010
Member. DuPont Northwestern Dow Collaborative Access Team Advisory Board. 2002 -
Scientific Advisory Committee. Northeast Collaborative Access Team, Advanced Photon Source, Argonne National Laboratory, 2007 - 2013
Steering Committee. APS Users Organization, 2008-2011

Board of Scientific Counselors, National Heart, Lung, and Blood Institute, National Institutes of Health, 2012 -

Publications (in chronological order)

1. Mondragón,A. and Ortega-Blake,I. Proton Transference in Hydrogen Bonded Systems. I. Uracil Tautomerism. *Int. J. Quantum Chem.*, **22**, 89-100, (1982).

2. Westhof,E., Altschuh,D., Moras,D., Bloomer,A.C., Mondragón, A., Klug,A. and Van Regenmortel,M.H.V. Correlation Between Segmental Mobility and the Location of Antigenic Determinants in Proteins. *Nature*, **311**, 123-126, (1984).

3. Mondragón,A., Bloomer,A.C., Gebhard,W. and Holmes,K.C. Preliminary Atomic Model for Tobacco Mosaic Virus. *Proceeding of the 16th FEBS Congress Part A*, 363-373, (1985).

4. Turner,D.R., Mondragón,A., Fairall,L. Bloomer,A.C., Finch,J.T., van Boom,J.H. and Butler,P.J.G. Oligonucleotide Binding to the Coat Protein Disk of Tobacco Mosaic Virus: Possible Steps in the Mechanism of Assembly. *Eur. J. Biochem*., **157**, 269-274, (1986).

5. Harrison,S.C., Anderson, J.E., Koudelka,G.B., Mondragón,A., Subbiah,S., Wharton,R.P., Wolberger,C. and Ptashne,M. Recognition of DNA Sequences by the Repressor of Bacteriophage 434. *Biophys. Chem*., **29**, 31-37, (1988).

6. Harrison,S.C., Ptashne,M., Anderson,J.E., Dong,Y.C., Drottar, M., Koudelka, G.B., Mondragón,A., Subbiah,S., Wharton,C. and Wolberger,C. Recognition of DNA Sequences by the Repressor and Cro Proteins of Bacteriophage 434. *Bull. Inst. Pasteur (Paris)*, **86**, 55-64, (1988).

7. Mondragón,A., Wolberger,C. and Harrison S.C. The Structure of Phage 434 Cro protein at 2.35 Å Resolution. *J. Mol. Biol.*, **205**, 179-188, (1989).

8. Mondragón,A., Subbiah,S., Almo,S.C., Drottar,M. and Harrison,S.C. The Structure of Phage 434 Repressor Amino Terminal Domain at 2.0X Resolution. *J. Mol. Biol.*, **205**, 189-200, (1989).

9. Mondragón,A. and Harrison,S.C. The Phage 434 Cro/OR1 Complex at 2.5 Å Resolution. *J. Mol. Biol.*, **219**, 321-334, (1991).

10. Lima, C.D., Wang,J.C. and Mondragón,A. Crystallization of a 67kDa Fragment of *Escherichia coli* DNA Topoisomerase I. *J. Mol. Biol.*, **232**, 1213-1216, (1993).

11. Lima,C.D., Wang,J.C. and Mondragón,A. Three-dimensional structure of the 67K N-terminal fragment of *E. coli* DNA topoisomerase I. *Nature*, **367**, 138-146, 1994.

12. Lima,C.D. and Mondragón,A. Mechanism of type II DNA topoisomerases: A tale of two gates. *Structure*, **2**, 559-560, 1994.

13. Sharma,A., Hanai,R. and Mondragón,A. Crystal Structure of the Amino Terminal Fragment of Vaccinia Virus DNA Topoisomerase I at 1.6 Å Resolution. *Structure*, **2**,767-777, 1994.

14. Sharma,A. and Mondragón,A. DNA topoisomerases. *Current Opinion in Structural Biology*, **5**, 39-57, 1995.

15. Mondragón,A. Unraveling transposition: γδ resolvase in complex with DNA. *Structure*, **3**, 755-758, 1995.

16. Lue,N., Sharma,A., Mondragón,A. and Wang,J.C. A 26 kDa yeast DNA topoisomerase I fragment: crystallographic structure and mechanistic implications. *Structure*, **3**, 1315-1322, 1995.
17. Mondragón,A. Solving the *cis/trans* paradox in the Int family of recombinases. *Nature Struct. Biol.*, **4**, 427-429, 1997.
18. Mondragón,A. and DiGate, R. Structure of *E. coli* DNA topoisomerase III, *Structure*, **7**,1373-1383,1999.
19. Grum, V.L., Li, D., MacDonald,R.I. and Mondragón, A. Structures of two repeats of spectrin suggest models of flexibility. *Cell*, **98**, 523-535, 1999.
20. Feinberg, H., Lima, C.D. and Mondragón, A. Conformational changes in *E. coli* DNA topoisomerase I. *Nature Struct. Biol.*, **6**, 918-922, 1999.
21. Feinberg, H., Changela, A. and Mondragón, A. Protein-nucleotide interactions in *E. coli* DNA topoisomerase I. *Nature Struct. Biol.*, **6**, 961-968, 1999.
22. Li, Z., Mondragón, A., Hiasa, H., Marians, K.J., and DiGate, R.J. Identification of a unique domain essential for *Escherichia coli* DNA topoisomerase III-catalysed decatenation of replication intermediates. *Mol. Microbiol.*, **35**, 888-895, 2000.
23. Li, Z., Mondragón,A, and DiGate, R. The mechanism of type IA topoisomerase-mediated DNA topological transformations. *Molecular Cell*, **7**, 301-307, 2001.
24. Changela, A., Ho, K.C., Shuman, S. and Mondragón, Structure and Mechanism of the RNA Triphosphatase Component of Mammalian mRNA Capping Enzyme. *EMBO J.*, **20**, 2575-2586, 2001.
25. Changela, A., DiGate, R. and Mondragón, A. Crystal structure of a complex of a type IA DNA topoisomerase with a single-stranded DNA molecule. *Nature*, **411**, 1077-1081, 2001.
26. Perry,K. and Mondragón, A. Biochemical Characterization of an Invariant Histidine Involved in *Escherichia coli* DNA Topoisomerase I Catalysis. *J. Biol. Chem.*, **277**, 13237-13245, 2002.
27. Krasilnikov AS, Yang X, Pan T, Mondragón A. 2003. Crystal structure of the specificity domain of ribonuclease P., *Nature*, 421. 760-764, 2003.
28. Changela, A. Perry, K., Taneja, B. and Mondragón, A. DNA manipulators caught in the act. *Current Opinion in Structural Biology*, **13**, 15-22, 2003.
29. Krasilnikov, A.S. and Mondragón, A. On the occurrence of the T-loop RNA folding motif in large RNA molecules. *RNA*, **9**, 640-643, 2003.
30. Changela A., Chen K., Xue Y., Holschen J., Outten C.E., O'Halloran T.V. and Mondragón A. Molecular basis of metal-ion selectivity and zeptomolar sensitivity by CueR. *Science*, **301**, 1383-1387, 2003.
31. Taneja B., Maar S., Shuvalova L., Collart F., Anderson W., Mondragón A. Structure of the bacillus subtilis YYCN protein: a putative N-acetyltransferase. *Proteins*, **53**, 950-952, 2003.
32. Perry K. and Mondragón A. Structure of a complex between E. coli DNA topoisomerase I and single-stranded DNA. *Structure*, **11**, 1349-1358, 2003.
33. Kusunoki, H., MacDonald, R.I., and Mondragón, A. Structural Insights into the Stability and Flexibility of Unusual Erythroid Spectrin Repeats. *Structure*, **12**, 645-656, 2004.

34. Krasilnikov, A.S., Xiao,Y., Pan,T. and Mondragón,A. Basis for Stuctural Diversity in Homologous RNAs, *Science*, **306**, 104-107, 2004.
35. Kusunoki, H., Minasov, G., MacDonald, R.I., and Mondragón, A. Independent Movement, Dimerization and Stability of Tandem Repeats of Chicken Brain α-Spectrin, *J. Mol. Biol.*, **344**, 495-511, 2004.
36. Changela A., Martins A., Shuman S., Mondragón A. Crystal structure of baculovirus RNA triphosphatase complexed with phosphate. *J. Biol. Chem*., **280**,17848-17856, 2005.
37. Mondragón, A. Unraveling the mechanistic details of topoisomerases. Structure, **13**, 502-503, 2005
38. Torres-Larios A., Swinger K. K., Krasilnikov A. S., Pan T., Mondragón A. Crystal structure of the RNA component of bacterial ribonuclease P. *Nature*. **437**, 584-587, 2005.
39. Patel, A. Shuman, S., and Mondragón, A. Crystal structure of bacterial type Ib DNA topoisomerase reveals a preassembled active site in the asbence of DNA. *J. Biol. Chem*. **281**, 6030-6037**,** 2006.
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51. Rajan, R., Taneja, B., Mondragón, A., Structures of Minimal Catalytic Fragments of Topoisomerase V Reveals Conformational Changes Relevant for DNA Binding, *Structure*, **18**, 829-838, 2010.
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54. Baker, N.M, Weigand, S., Maar-Mathias, S, and Mondragón, A. Solution structure of DNA bound gyrase. *Nucleic Acids Research*, **39**, 755-766, 2011.
55. Reiter, Chan,C., Mondragón A. Emerging structural themes in large RNA molecules. *Curr. Opin. Struct. Biol.*, **21**, 319-326, 2011.
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60. Structure and function of the T-loop structural motif in non-coding RNAs. Chan,C.W., Chetnani,B, and Mondragón, A. WIRE RNA, 2013, in press.
61. Structural Studies of RNase P. Mondragón, A. *Annu. Rev. Biophysics*, 2013, in press.
62. Studies of bacterial topoisomerases I and III at the single-molecule level. Terekhova K., Marko J.F., and Mondragón A. *Biochem Soc Trans.*, **41,** 571-575, 2013. In press.